



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9709/32

June 2026

1 hour 50 minutes

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

- 1 (a) Sketch the graph of $y = |3x - 2a|$, where a is a positive constant.

[1]

- (b) Hence or otherwise solve the inequality $|3x - 2a| < x + 5a$.

[3]

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[4]

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[4]

[illegible]

[4]

[illegible]

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- 5 The polynomial $2x^4 + ax^3 + 4x^2 + bx - 3$ is denoted by $p(x)$.

It is given that $(x^2 + x + 1)$ is a factor of $p(x)$.

- (a) Find the values of a and b .

[4]

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- (b) Hence, show that $(x + 3)$ is a factor of $p(x)$.

[2]

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- 6 (a) Sketch the graph of $y = |x + 3a|$, where a is a positive constant.

[1]

- (b) Hence or otherwise solve the inequality $|x + 3a| > a - 2x$.

[2]

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- 11 Solve the equation $\frac{e^x + 2e^{-x}}{e^x - 3} = 4$. Give your answer correct to 3 decimal places.

[5]

[6]

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13 Solve the equation

$$\ln(1 - e^{-2x}) + 3 = 0.$$

Give your final answer correct to 4 decimal places.

[3]

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[3]

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[2]

This image shows a blank sheet of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

15 The variables x and y satisfy the equation $Ay = b^x$, where A and b are constants.

(a) Show that the graph of $\ln y$ against x is a straight line. [2]

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(b) When $x = 3.4$, $\ln y = 0.86$ and when $x = 5.7$, $\ln y = 2.56$.

Find the value of A and the value of b . Give your answers correct to 2 significant figures. [3]

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- This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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- This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

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